

Climate Resilience: A Multi-Source Analysis of Climate Policy Evolution (2013–2025)

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This report documents a data mining pipeline bringing together multiple separate sources of climate data. The exploration analyzes the evolution of climate strategies since the Paris agreement (2015), with a particular focus on the integration of adaptation measures alongside mitigation efforts in a select group of countries.

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1 Introduction

Climate policy conversations have historically focused on the mitigation of climate change. However, the increasing frequency and severity of climate impacts has increased the importance of adaptation measures in national policy portfolios. The Paris Agreement (2015) marked a pivotal moment in global climate governance, establishing a framework for both mitigation and adaptation efforts.

Importantly, international climate databases systematically under represent adaptation policies, particularly in the Global South where adaptation is a *survival imperative*. When adaptation is invisible in the data, it becomes invisible in policy debates, climate finance allocation and accountability mechanisms. The lack of comparative global analysis and systematic scrutiny of national adaptation legislation and policies has “hampered the ability to assess progress, identify gaps, and understand effective frameworks” [1]. This project serves as a digital audit of climate ambition across the selection of countries with an aim to discover if and how countries are shifting towards adaptation adoption.

The comparative analysis of the Global North and Global South is critical to understanding the equity implications of climate policy evolution. The Global North is responsible for the majority of emissions [5] and has more resources to invest in both mitigation and adaptation. Those in the Global South, whilst being less responsible for the acceleration of climate change, are the ones who bear the brunt of it [4].

2 Research Question

Has there been a shift towards climate adaptation adoption in policy portfolios after the Paris Agreement, and to what degree does this shift differ between Global North and Global South countries?

Key Terms:

- **Climate Policy Portfolio:** The collection of all climate-related policy strategies enacted by a country.
- **Mitigation:** Efforts to reduce or prevent the emission of greenhouse gases to limit global warming.
- **Adaptation:** Efforts to adjust to actual or expected climate change and its effects.

Note: The terms “Global North” and “Global South” are used as broad categorizations to capture the systemic inequalities in climate responsibility and vulnerability.

Boundaries:

Temporal Boundary (2013–2025)

The analysis will focus on the period from 2013 to 2025 in order to capture the policy landscape leading up to the Paris Agreement (2015) and the subsequent years. This temporal boundary provides a clear before-and-after comparison to assess the impact of the Paris Agreement on climate policy evolution.

Country Selection

The selection of 13 countries (Figure 1) is designed to provide a diverse sample that represents different levels of vulnerability, responsibility, and geographic distribution.

Selected Countries and their GHG Emissions (since 2013)

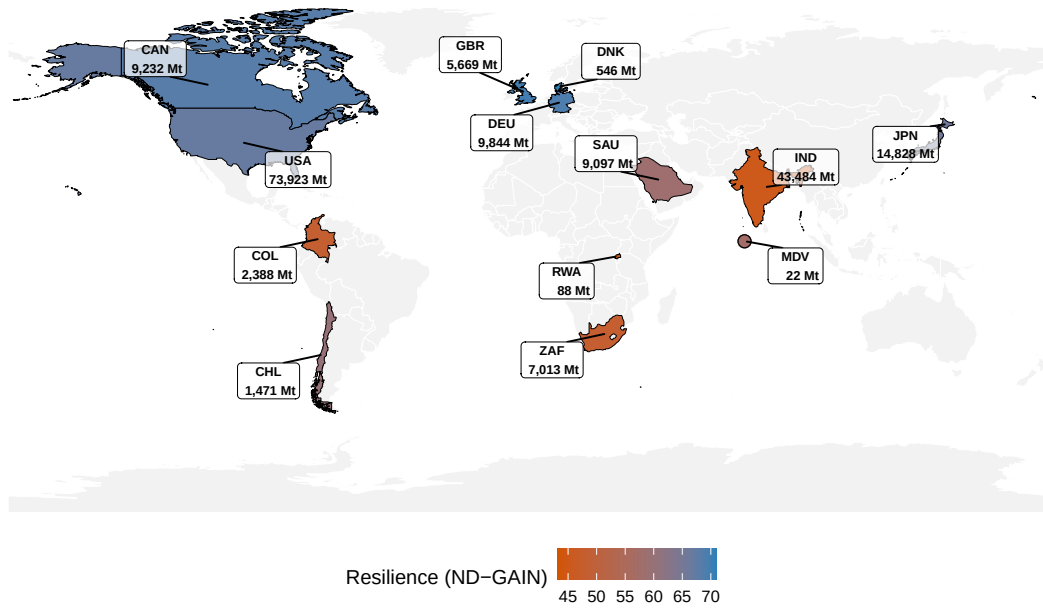


Figure 1: Selected Countries and their Emissions (Metric Tons)

Note: The United States officially left the Paris Agreement (for the second time) on January 27 2026”

3 Data Sources

3.1 NewClimate Institute’s Climate Policy Database (CPDB) [2]

The CPDB provides the baseline for policy adoption with broad geographic coverage. The CPDB’s provides structured policy data with policy types (mitigation, adaptation), decision dates and country codes, allowing for a quantitative analysis of policy trends over time. Its primary limitation is a “Global North bias,” where data coverage is stronger (This is a gap addressed by the inclusion of NDC texts).

The cleaning and processing of the CPDB data is explained in Github issue #7.

3.2 UNFCCC Nationally Determined Contributions (NDCs) [3]

These documents serve as primary statements of government priority. While they allow for a nuanced NLP analysis of “Lexical Hardness” (Hard vs. Soft law) as well as “Adaptation focus” (Percentage of text focused on adaptation), they represent diplomatic pledges rather than tangible enforcement. The plus side of using NDCs is that each country is able to represent themselves as they choose, removing the reliance on third-party data collection and coding which may be biased or incomplete, particularly for the Global South.

The collection and processing of the NDCs is explained in issue #8.

3.3 ND-GAIN Index (Resilience Benchmarking) [4]

The ND-GAIN Index provides the context of necessity. The ND-GAIN scores are a composition of vulnerability and readiness indicators. While the Vulnerability score measures physical exposure, the overall Index balances this against a country’s Readiness (its economic, social, and governmental capacity to deploy adaptation solutions). The overall index is therefore a measure of resilience.

The tidying of nd-gain is explained in issue #12.

3.4 World Bank World Development Indicators (WDI) [5]

The WDI provides critical contextual data on emissions responsibility (historical cumulative emissions).

The accessing and tidying of WDI is explained in issue #16.

4 Technological Approach & Challenges

The entire workflow is orchestrated by a single master script (`main.R`) that manages:

1. **Dependency installation**
2. **Data acquisition**
3. **Data processing** (cleaning, feature engineering).
4. **Analysis**
5. **Output generation** (figures, tables, this report).
6. **Environment Cleaning** (removing temporary files and data).

Data Acquisition

To implement a multi source triangulation strategy I first bridged R with Python (reticulate) to access the CPDB API, then developed a custom scraping pipeline for UNFCCC NDCs. This involved automated PDF downloading and text extraction via pdftools. The context data from ND-GAIN and World Bank WDI was simpler to access and clean but provide critical context for the analysis.

Key challenge:

I had initially planned to focus my analysis on the OECD CAPMF database which provides stringency scores for climate policies. This is an extremely detailed source of data which required extensive cleaning in order to create a structured data frame. However I subsequently realized that the CAPMF data only covers a subset of the countries in my sample and mainly covers mitigation policies, so I made the decision to shift my focus towards the CPDB which has a broader coverage of both countries and policy types.

Discussed in Issue #2

Adaptation tracking

To track the evolution of adaptation policies in the CPDB I developed a Keyword-Based Heuristic to classify policies as “adaptation”, “mitigation” or “both” (based on a more complex policy intent label included in the data set). It is important to note that for my analysis I have looked at any policy that includes adaptation even if it also includes mitigation. This is because I am interested in the shift towards adaptation, so I want to capture any policy that includes adaptation in its language. The CPDB data has a bias towards Global North countries, so I have focused on relative ratios instead of absolute counts in order to get a clearer picture of the adaptation focus of each country. *Discussed in issue #1*

Lexical Audit & Feature Engineering

NDC Strategic intent was quantified through a Keyword-Based Heuristic. I developed two distinct lexicons:

- Focus Score: A ratio of adaptation-to-mitigation terminology normalized by document length.
- Ambition Score: A “Lexical Hardness” index comparing mandatory (“legally”, “must”) vs. aspirational (“aim”, “seek”, “aspire”) language, utilizing a smoothing constant (+0.001) to handle zero-count denominators in shorter documents.

These are discussed in issue #15, #5 and #4.

5 Results

5.1 Global Adaptation Growth

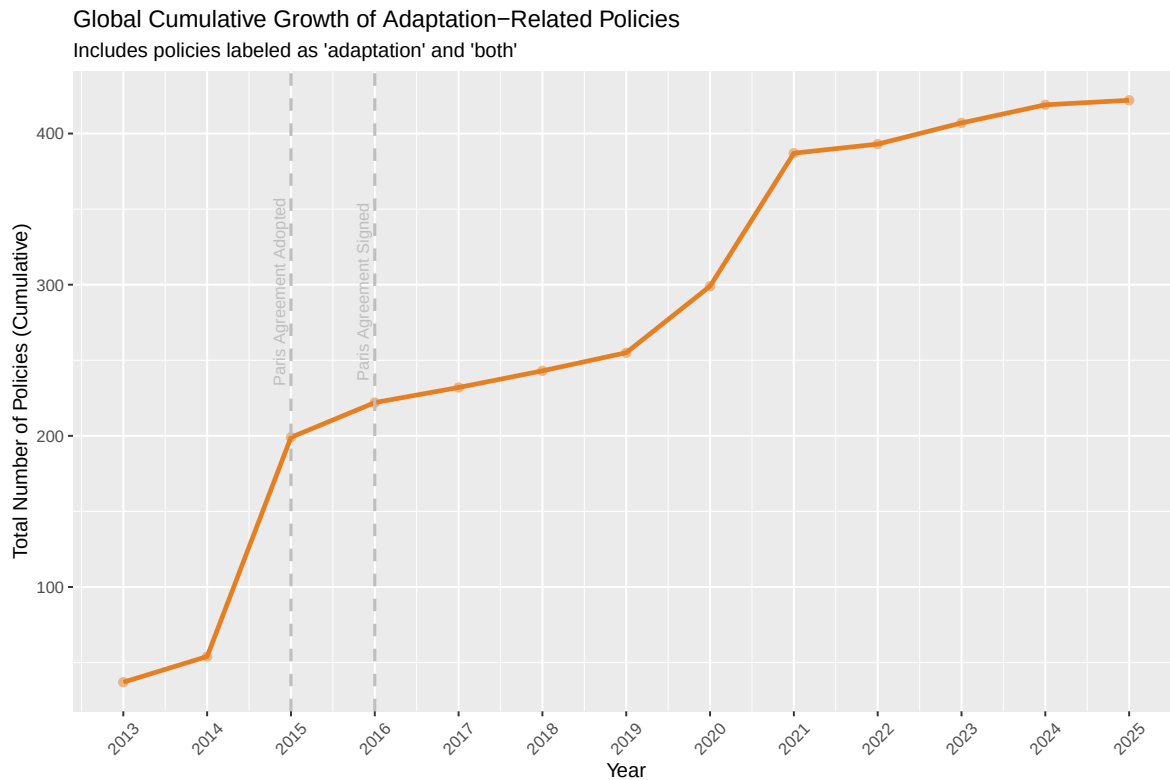


Figure 2: Global Cumulative Growth of Adaptation Policies (2013-2025) (CPDB)

Initial analysis of climate policies in the CPDB reveals a visible acceleration in global adaptation policy adoption starting around 2015 (the same year as the Paris Agreement). This is the starting point for our analysis, confirming that the Paris Agreement did indeed catalyze a shift in policy focus.

5.2 Policy Implementation per Country (The Adaptation Mix)

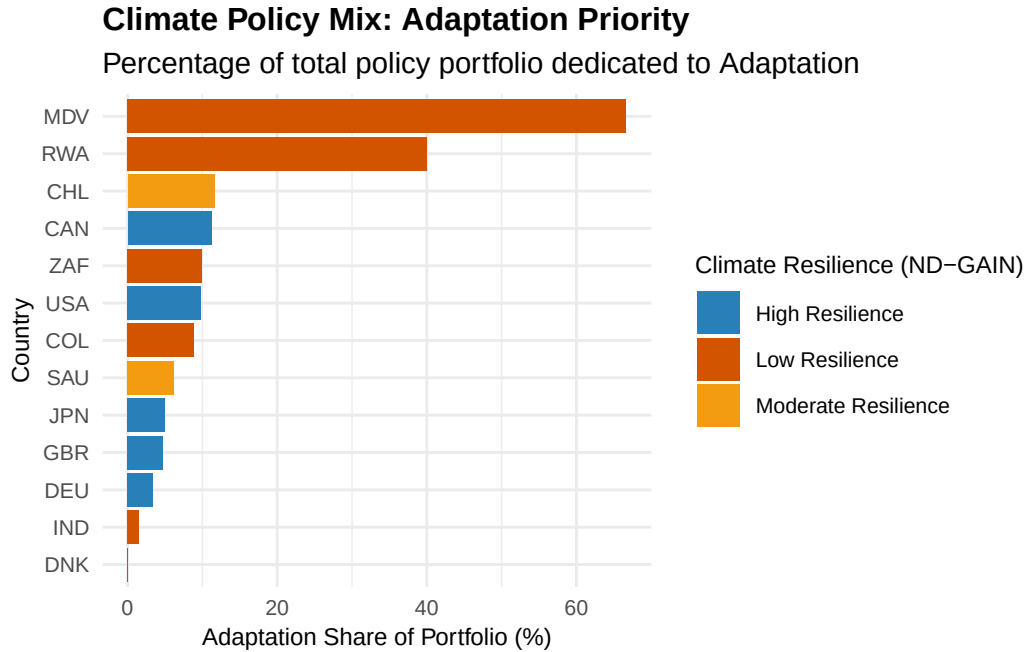


Figure 3: The Adaptation Mix: Percentage of Portfolio focusing on Adaptation (CPDB)

Zooming in to our chosen list of countries (Figure 3) we are able to see the spread of each policy portfolio. The ND-GAIN scores have been added in order to visualize the comparison of vulnerability. While some vulnerable nations like Rwanda show a high adaptation ratio, others like India lag behind. This would suggest that vulnerability may not be a driving factor in adaptation policy adoption.

However, it is important to keep in mind that this data set has its limitations (Global North bias, temporal lag, adaptation tracking difficulty) which may be skewing the analysis.

5.3 Lexical Adaptation Focus in NDC text (Responsibility Gap)

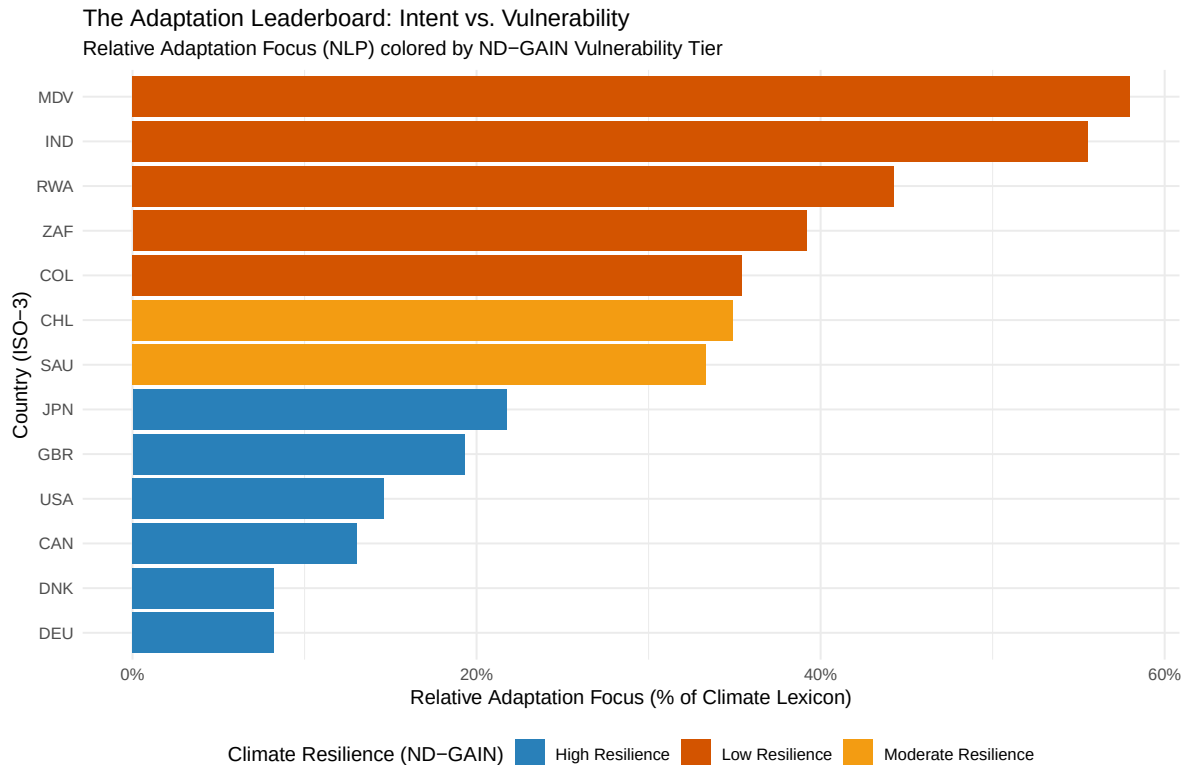


Figure 4: Adaptation Focus in NDC text

By shifting to an NLP-based analysis of each Nationally Determined Contribution (NDC) (Figure 5), the correlation between resilience and strategic intent becomes obvious. High-vulnerability countries dedicate significantly more “mind share” to adaptation in their NDCs compared to the Global North. This is the Responsibility Gap in action: the nations least responsible for historical emissions are the ones forced to prioritize survival strategies.

Comparing Policy Implementation (CPDB) to Strategic Focus (NDC)			
ISO-3	CPDB Mix (%)	NDC Focus (%)	The Gap (diff)
MDV	66.7	58.0	-8.7
IND	1.6	55.6	54.0
RWA	40.0	44.3	4.3
ZAF	10.0	39.2	29.2
COL	8.9	35.4	26.5
CHL	11.6	34.9	23.3
SAU	6.2	33.3	27.1
JPN	5.0	21.7	16.7
GBR	4.7	19.3	14.6
USA	9.8	14.6	4.8
CAN	11.2	13.0	1.8
DEU	3.4	8.2	4.8
DNK	0.0	8.2	8.2

Figure 5: The Commitment Gap

Figure 5 shows a comparison between the CPDB policy adaptation mix and the NDC lexical adaptation focus. The gap between intent and policy implementation seems to be particularly wide for the Global South. This is the Commitment Gap: vulnerable countries are prioritizing adaptation in their strategic documents but are not always able to translate this into binding policies. This is likely due to resource constraints as adaptation policies often require significant investment and international finance is often conditional on the availability of domestic resources [1].

5.4 Adaptation Focus vs. Ambition/Stringency in NDCs (Commitment Gap)

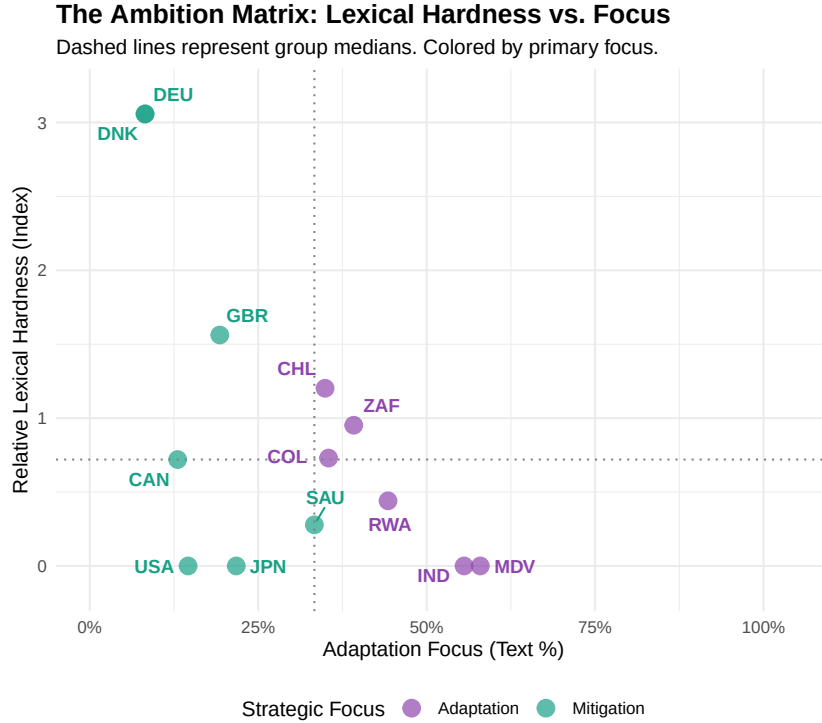


Figure 6: Climate Ambition Matrix: Lexical Hardness vs. Adaptation Focus

The Ambition Score reflects the relative frequency of mandatory language (e.g., shall, must, mandate) versus aspirational terms (e.g., should, aim, aspire), normalized against the group mean. While lexical hardness varies across all regions, the strategic focus remains deeply polarized. By centering the matrix on group medians, the split is visually stark: the Global South and the Global North are legislating for fundamentally different realities.

6 Discussion

Has there been a shift towards climate adaptation adoption in policy portfolios after the Paris Agreement, and to what extent does this shift differ between Global North and Global South countries?

There has been a noticeable shift towards climate adaptation adoption in policy portfolios since the Paris Agreement and this shift differs significantly between Global North and Global South countries. The Global South, which is more vulnerable to climate impacts, shows a higher focus on adaptation in their NDCs but faces significant challenges in translating this into binding policies, as evidenced by the Commitment Gap. The Global North, while having more resources, shows less focus on adaptation both in their NDCs and in actual policy implementation.

The Responsibility Gap and the Commitment Gap both illustrate the injustice at the heart of climate change: those who are least responsible for causing it are the ones who suffer its worst effects and have the greatest need for adaptation, yet they have the least capacity to do so.

The Commitment Gap finding has particular significance. It demonstrates that Global South adaptation priorities are substantive (evidenced by high keyword frequency), but these priorities are constrained by resource limitations (manifested as soft law language because commitments are conditional on international finance [1]).

7 Limitations

There are a great deal of limitations within this body of work, many of which are related to data reliability and coverage as well as simplified heuristics for tracking adaptation focus and strategic intent.

The Global North bias of CPDB means that the adaptation policies of many Global South countries may be underrepresented, potentially skewing the analysis. The use of relative ratios instead of exact counts is an effort to avoid this bias. The temporal lag of policy data is another concern, as there may be a delay between the strategic intent expressed in NDCs and the actual implementation of policies.

The NDC analysis is limited due to the variability in language and the challenge of quantifying qualitative commitments. The execution of the Lexical Audit was constrained by the depth of the text-mining heuristics. The current Keyword-Based Heuristic assumes that word frequency correlates with political priority. While normalized for document length, this does not account for the context of mentions *Discussed in issue #5 and issue #4*.

Future iterations would benefit from a more robust NLP framework to move beyond simple frequency counts and more accurately distinguish between aspirational rhetoric and enforceable domestic mandates. An interesting additional angle would be to study the impacts of this shift towards adaptation on actual climate outcomes, such as changes in vulnerability or resilience

metrics over time as well as climate change impacts. Work could also be done to explore how those responsible for the greater share of emissions (the Global North) can be held accountable for the adaptation needs of the Global South, particularly in the context of international climate finance and loss and damage discussions.

8 References

- [1] Grantham Research Institute on Climate Change and the Environment. (2025). *Climate change adaptation laws and policies: Global trends and insights 2025*. London School of Economics and Political Science. <https://www.lse.ac.uk/granthaminstitute/wp-content/uploads/2026/02/Climate-change-adaptation-laws-and-policies.pdf>
- [2] NewClimate Institute, Wageningen University and Research, & PBL Netherlands Environmental Assessment Agency. (2024). *Climate Policy Database* [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.7774109>
- [3] United Nations Framework Convention on Climate Change. (2026). *NDC Registry* [Database]. <https://unfccc.int/NDCREG>
- [4] University of Notre Dame. (2026). *Notre Dame Global Adaptation Initiative Country Index* [Data set]. <https://gain.nd.edu/our-work/country-index/>
- [5] The World Bank. (2026). *World Development Indicators* [Data set]. <https://databank.world-bank.org/source/world-development-indicators>

9 LLM Declaration

The following LLMs have been used in the process of this project:

- Gemini for technical troubleshooting, code review and idea exploration: I used Gemini throughout the coding process and also as a way to talk through my ideas. All code has been a joint effort, never a complete copy-paste but more of a back and forth discussion about best methods. I have a background in software development but have never used R before, so Gemini has helped me navigate the syntax.
- Github Copilot for code suggestions and auto completion: I used Copilot to speed up coding and get suggestions. Again, all code has been a joint effort, with me writing the logic and Copilot suggesting syntax and functions.
- Scispace for help writing this report: I used Scispace to help organize my thoughts and ensure that I am covering all the necessary sections and points.

Total Word Count (minus LLM Declaration): 2226